Algae class presenter notes

INSTRUCTOR EDITION · English review · 8 November 2026

Use these notes to prepare the explanation behind the visual presentation. Timings refer to a 60-minute session. The student guide carries the deeper derivations and references. This is Stage 2: the public deck has not yet been corrected or translated.

Time	Slides	Teaching purpose
00 to 12 min	1 to 3	Commuting poll and life-cycle boundary
12 to 20 min	4 to 7	Energy history, energy mix and travel demand
20 to 23 min	8	Electric-car life cycles and Brazilian context
23 to 28 min	9 to 10	Mass constraints and fleet turnover
28 to 33 min	11 to 12	SAF and the chemistry of fuel routes
33 to 38 min	13 to 15	Algae bioprocesses and material definitions
38 to 50 min	16 to 19	Your research and process interpretation
50 to 60 min	20 to 21	Algae Farm and debrief

Preparation before the session

Open the public presentation and the instructor poll link on the presentation computer. Start or select the intended fresh collection using the existing controls and confirm that its QR points to that collection. Check the internet connection, projector readability, game load and sound if a video is used. Keep a copy of the worked commuting example for a fallback calculation.

The student guide's numbers are the audited ones. The current energy slide still needs the 80.9% correction during Stage 3. Notes in the added research deck about Creglicerase are unrelated and should not be read. Four pond maxima remain unverified and are excluded from the teaching facts here.

For the current English projection, explain terms orally in the class's teaching language. Stage 3 will make the presentation, notes, guide and Anki consistently Portuguese while retaining the existing Portuguese game link.

Opening activity and commuting calculation

00 to 12 min · Slides 1 to 3

Prompt and collection

Suggested opening: “Before we discuss a new fuel, let us describe a transport service we already use. How do we get to campus during a typical week, and what has to be counted to estimate its impact?” Give the QR instructions immediately. Avoid asking for addresses or names.

Explain that ida and volta are separate journeys. A week can contain ten journeys with different outward and return modes; up to 14 are allowed. Enter one approximate one-way distance, then allocate the journeys across modes. “Carona +1” means two occupants including the driver. Mention that the shared distance cannot describe every mixed route accurately.

Build one calculation together

Ask students which quantity the emission factor must multiply. Use Ana’s example only if useful: fourteen 8-km journeys, seven bus and seven bicycle. Ask for the distance by mode before showing any emissions. The answer is 56 passenger-km for each mode, not 112 for each and not an additional round-trip multiplier.

$$56 \times 0.12242 + 56 \times 0.00017 = 6.86504 \ \mathrm{kg\ CO_2e/week}$$

Ask how the class total and mean are calculated. Sum personal weekly estimates; divide by response count for the mean per respondent. Distinguish this from a mean per journey. Explain that unknown-mode distances make the total partial. Use response count and distance coverage alongside the total.

Interpret the result before ranking people

The reference is ADEME France, including manufacture and fuel-chain use, with no extra food or infrastructure. It is a scenario, not a measurement of each student’s Brazilian vehicle. Uber is a reference car with one paying passenger and no empty-running allowance. A bus factor already includes average occupancy. [R05–R07]

If showing the flight comparison, call it the same passenger-distance under a selected flight profile. The result includes contrail effects and is CO_2e, not exhaust CO_2 alone. Ana’s short-flight equivalent is 25.2 kg, about 3.67 times her scenario. Do not promise a universal multiplier. [R32]

Transition

“We can do the multiplication quickly. Choosing a fair boundary is the scientific part. Now let us see why modern society depends so strongly on fossil energy.”

Energy history transport demand and electric cars

12 to 23 min · Slides 4 to 8

The fossil graph

Ask students to locate a parent's or grandparent's lifetime on the graph. State its actual 1800–2023 range and TWh unit. It shows fossil primary energy, not all energy over human history. The family-history prompt is a discussion device, not another dataset. [R03]

The energy mix

Use the corrected supplied values: 27.6% coal + 30.2% oil + 23.1% gas = 80.9% fossil in world total energy supply, 2022. The prepared slide currently says 81.7%; Stage 3 must correct it. If the old slide is used before then, state the correction explicitly rather than repeating the error. "Other" is not equivalent to renewables or zero emissions. [R01]

Ask whether an electricity statistic answers the same question. Use Brazil's 86.8% renewable electricity supply in 2025 as a separate example of a different denominator. Do not call it the renewable share of all energy or a carbon factor. [R11]

Transport shares and urban options

The 2018 modal chart assigns 74.5% of transport CO_2 to road, 11.6% to aviation and 10.6% to shipping. State the year and denominator. Ask why a smaller sector can still deserve research: climate relevance is not the only determinant of substitution difficulty. [R04]

For city transport, connect walking, cycling, sharing and public transport to actual services. Avoid presenting a choice as morally available to everyone regardless of distance, disability or safety. The teaching task is to identify changes to a transport system as well as changes to a vehicle.

Electric cars

The current EU bars are 235, 63 and 52 g CO_2e/km in an ICCT model for medium cars sold in 2025. Do not state that these are Brazilian measurements. The Brazil-specific study gives 155 versus 55 for compact flex-fuel/BEV cases, with lifetime fuel and electricity assumptions. [R09, R10]

Use a matched comparison to explain manufacture plus use. A bigger vehicle can require more materials and energy; that does not negate the modelled BEV benefit within the same class. Ask what changes if electricity, lifetime mileage or vehicle size changes. Skip a universal break-even distance because it depends on those inputs.

Transition

"Electricity is a strong option for many journeys. The next constraint is how much energy a vehicle must carry and for how long."

Onboard energy and aviation fuel routes

23 to 33 min · Slides 9 to 12

Make the units do the explanation

Ask which unit is required before comparing jet fuel and batteries. Convert 250 Wh/kg to 0.9 MJ/kg. Jet A at 43.2 MJ/kg gives a stored-energy ratio of 48. Label that as stored energy per mass. A comparison of range needs propulsion efficiency, aircraft design, reserves and battery system mass. [R13, R14]

If a student asks whether motor efficiency changes the conclusion, show the hypothetical useful-energy example: 43.2 × 0.35 versus 0.9 × 0.90 gives a ratio near 18.7. Explain that the efficiencies are invented sensitivity inputs, not a model of a particular aircraft. Acknowledge charging and short-route applications rather than ruling out every electric plane or ship.

Fleet turnover

Use 25–30-year passenger retirement and ship technical-life ranges as evidence of inertia. They describe different metrics and do not fix a single replacement date. Ships can be modified, aircraft designs differ, and fuel availability also matters. Rail supplied by overhead electricity belongs to a different technical case. [R15, R16]

Define SAF precisely

Separate technical qualification from sustainability. Combustion still releases CO_2; the claimed reduction comes from the assessed life cycle. Avoid assigning one saving percentage to all SAF or inferring universal approval from a 100%-SAF engine-test photo. ICAO's online route table includes a dated narrative, so do not present its route count as a newly verified 2026 total. [R17, R18]

Use the chemical distinction

Ask students whether FAME molecules are the same as hydrocarbon kerosene. Explain the ester group and the need for a different conversion route. Hydroprocessing uses hydrogen and removes oxygen; ethanol-to-jet requires conversion of small alcohol molecules into a suitable hydrocarbon mixture. HTL first makes a biocrude intermediate that needs upgrading. [R17, R20, R21]

Use LanzaJet's November 2025 announcement as a dated, company-reported milestone. Contrast its evidence type with the PNNL experimental/TEA study without implying algae HTL is already certified commercial supply. A micrograph on a general route slide may be illustrative rather than the strain used in the cited experiment. [R19, R20]

Transition

"We need both useful molecules and a process that makes them reliably. This brings us from the fuel route to the organism and the laboratory."

Algae hosts materials and process boundaries

33 to 38 min · Slides 13 to 15

Narrow the biological scope

Show the diversity collage briefly, then identify the research organism as a green microalga. A seaweed image or cyanobacterium should not silently stand in for C. pacifica. Ask what a photoautotrophic culture needs beyond CO_2 and sunlight: water, nutrients, circulation, thermal management and a way to harvest useful material.

A biochemical engineering class can follow the carbon balance directly. One kilogram of dry biomass assumed to contain 50% carbon corresponds to 1.83 kg CO_2 through 44/12. Label the carbon fraction as an example. Uptake is not a full avoided-emissions calculation, because carbon can return during combustion or degradation and the process has other inputs.

Follow the process

Use strain → cultivation → harvest → recovery → conversion → qualified product. Ask which measurements belong at each step. Optical density, dry weight, lipid fraction and recovered fuel mass answer different questions. The guide's mass-balance example provides the deeper explanation; there is no need to put all its text on a slide.

Explain what the material photograph establishes

A footwear or surfboard image establishes an application example, not a biobased fraction or composting result. Define biobased as origin, biodegradable as conditional breakdown, and compostable through its relevant conditions and criteria. For the tested 2020 foam, 12-week mass loss differed between soil and compost; do not generalize it to every algal polymer. [R22, R29]

Optional short story

C. pacifica came from a San Diego soil sample. Ask whether students expected a microalga with a Pacific-sounding name to come from the ocean. Then explain how tolerance studies can inform host choice while still needing production data. Do not imply a habitat name proves outdoor process performance. [R24]

Transition

"The next figures show the part we can test directly: select a host, change a biological target, measure the phenotype and then recover a product."

Explaining your research figures

38 to 50 min · Slides 16 to 19

The research question

Frame the ACS study as a pilot-scale pipeline for engineering a resilient host toward useful biomass composition and product conversion. The article appeared online in November 2024 and in the 2025 journal volume. Use the co-first-author attribution accurately. Do not describe it as a demonstration of qualified SAF or a complete commercial LCA. [R23]

Selecting and measuring a phenotype

The selection evidence supports improved high-light tolerance. Do not broaden that to a proven gain in every environmental tolerance. For a deeper explanation, distinguish genetic change from the phenotype and from process performance. A laboratory phenotype can be useful without establishing annual cultivation reliability.

Before showing the composition bars, ask students to identify the denominator. The unit is grams per 100 g dry biomass. Dof lipid content changes 28.99 to 37.00; PGM1 starch changes 36.08 to 45.83 in a separate strain. Ask for the difference first, then the relative increase. Answers: 8.01 percentage points and 27.63% relative lipid increase. [R23]

Connect content to productivity

Ask what else is needed to calculate fuel per area per day. Elicit biomass productivity, recovered fraction and conversion yield. Use the guide's invented 10-g, 30%-fraction scenario: 10 × 0.30 × 0.80 × 0.90 = 2.16. Explain why increased composition can be offset by reduced growth. Keep simultaneous measurements and experiment identity together.

Optional replacement figure from the new deck

Use one carefully explained TF histogram or one enzyme-assay panel if research depth is needed. This should replace time elsewhere, not add fourteen slides. For cytometry, identify Nile Red, PE-A, biological replicates and fluorescence units. For enzymes, identify substrate and assay temperature; distinguish the PU dispersion from the heated PET experiment. [R25, R26]

The proposed lab narrative is candidate choice → construct/strain verification → matched culture experiment → phenotype assay → chemical product analysis. State the measurement actually reported at each step. Do not invent laboratory steps, productivity gains or commercial validation beyond the source.

Research story that invites critical thinking

The ACS paper reports initial PGM1 growth trouble followed by recovery in later rounds. Ask which controls and repeat runs help distinguish inoculum problems, contamination and a strain effect. The reported suggested causes are not a proven diagnosis. [R23]

Conclude the section by asking what evidence is still needed for "better than petroleum." This sets up the game as a process decision exercise rather than a competition to prove a predetermined result.

Running and debriefing Algae Farm

50 to 60 min · Slides 20 to 21

Minute	Action
50 to 51	Open the embedded Portuguese game and explain the objective shown.
51 to 57	Allow six minutes of play. Ask students to remember one decision and its consequence.
57 to 60	Debrief the limiting resource, recovered product and missing real-world data.

Instructions to give aloud

“Treat your first choice as a hypothesis. Keep track of what you changed, what useful output you obtained and which resource became limiting. At the end, we will ask whether the strategy would still look good with a complete process boundary.”

Use the already-linked Portuguese game. The presentation’s existing game URL includes a six-minute scenario. If students instead open the standalone game, confirm its displayed settings rather than assuming the parameters were preserved.

Three questions for the last discussion

What actually limited your result? Which measurement would help test that explanation? How would energy, recovery losses and product lifetime change the comparison? Encourage a specific variable or measurement rather than accepting “more sustainable” as a sufficient answer.

Compare equivalent objectives. Different contracts, starting resources or scenarios can produce different score scales. Recovered product per relevant resource can be more informative than the raw number of game tokens. The game rules remain fictional teaching assumptions, not an audited carbon model or fuel certification. [R37]

Final statement

“Algae offer a way to make useful carbon-containing products and enzymes. Engineering decides whether that biological possibility becomes a process with useful output and acceptable impacts. The evidence has to connect the organism, the conversion and the service it provides.”

If the internet fails

Use the printed Ana and Bruno example to complete the footprint discussion. For the final activity, ask small groups to choose between the three invented productivity scenarios in the guide and defend a boundary and one missing measurement. This preserves the intended reasoning activity without claiming that a nonfunctioning game ran successfully.

Presentation: https://molino-algae.github.io/algae-sustainable-future/

Game: https://molino-algae.github.io/justlab-site/games/algae-farm/?minutos=6&modo=direto&cenario=fixo

Questions students may ask

Are biofuels carbon neutral

Not by feedstock identity alone. Count cultivation, conversion, hydrogen or energy, transport and relevant land-use effects. Explain how uptake and later release are treated in the selected life-cycle method. A short carbon cycle does not remove process emissions. [R18]

Why is the bicycle factor so small

The selected 0.00017-kg/km constant is in the frozen ADEME source. Its upstream lifetime/manufacturing derivation was not reconstructed in Stage 1. Use it transparently as a database scenario and compare the separate ECF boundary rather than claiming one universal bicycle value. [R05, R08]

Does an electric car always beat every alternative

The cited studies compare matched cars under specified lifetime assumptions. They support a substantial BEV benefit within those scenarios; they do not establish a universal ranking of every vehicle, bus occupancy, trip length and electricity source. [R09, R10]

Does a brighter lipid image mean more biodiesel

It supports a fluorescence phenotype under the assay. Chemical analysis, biomass output and conversion recovery are still required to quantify useful product per volume, area or time. [R25]

Can growing algae digest PET in a pond

The cited PET test used separated, concentrated supernatant at 70°C. Whole-culture PU observations concern another substrate and evidence level. Avoid substituting that result for a replicated outdoor PET-recycling process. [R26]

Why not report the pond maximum of 16

That number belongs to a different preprint experiment in the added deck. Its maximum, time basis, dry/ash-free definition and replicates were not independently verified. Do not combine it with the ACS composition values or present it as annual commercial performance. [R27, R28]

How should uncertainty be explained

Keep measurement error, variation among cultures, temporal variation and model uncertainty distinct. The guide shows some central values for learning arithmetic. The original figures remain the authority for error bars, sample sizes and statistical tests. A significant phenotype change does not validate an entire process model.

Reference codes throughout these notes use the same ledger as the student guide and Stage 1 audit. The final reference page links the principal sources used during teaching.

References

Reference codes match the Stage 1 evidence audit. Historical scenarios retain their dates. Original research and official methods are linked below. Full source metadata accompanies this package.

R01 Supplied Sustainable Future course: source index

Supplied Sustainable Future teaching collection. Lecture 1, PDF p. 9 provides the historical 2022 world energy snapshot. Other relevant figures are indexed in the Stage 1 audit.

R03 OWID: Fossil fuels

Global fossil fuel consumption chart; Energy Institute and Smil underlying data.

Read the original source

R04 OWID: CO_2 emissions from transport

2018 modal breakdown; underlying IEA / ICCT.

Read the original source

R05 ADEME / Impact CO_2: frozen transport data

Archived source file; downloaded and checked against calculator constants.

Read the original source

R06 ADEME: gasoline compact/medium car

170 g CO_2e per passenger-km displayed, rounded; manufacturing and fuel breakdown.

Read the original source

R07 ADEME: thermal bus

122 g CO_2e per passenger-km displayed, rounded.

Read the original source

R08 Blondel, Mispelon & Ferguson (2011), Cycle more Often 2 cool down the planet!

Printed pp. 9–11; bicycle lifetime, manufacture/maintenance and dietary assumptions.

Read the original source

R09 ICCT (2025): Life-cycle GHG emissions of passenger cars in the EU

Summary comparison: medium cars sold in 2025; lifetime electricity 2025–2044.

Read the original source

R10 Mera et al. / ICCT (2023): Passenger-car life-cycle GHG emissions in Brazil

Printed pp. 15–17, Figs. 5–7; 22-year life, 288,000 km; electricity/fuels 2023–2044.

Read the original source

R11 EPE (2026): BEN 2026, base year 2025

Printed p. 35: 86.8% renewable share of domestic electricity supply (OIEE), including net imports.

Read the original source

References continued

R13 Boeing CASCADE documentation: energy and cost parameters

Table 58: Jet A lower heating value 43.2 MJ/kg; kWh-to-MJ conversion.

Read the original source

R14 Embry-Riddle: Electric aircraft

Battery pack specific energy about 0.20–0.25 kWh/kg; distinguish pack and cell.

Read the original source

R15 EASA: Sustainability in the aircraft end-of-life phase

Passenger aircraft typically retired after 25–30 years; cargo often 30–40.

Read the original source

R16 IRENA (2021): A pathway to decarbonise the shipping sector by 2050

Printed p. 25: technical ship lifetime usually 25–30 years.

Read the original source

R17 ICAO: SAF conversion processes

Approved Conversion Processes table; narrative explicitly dated July 2023.

Read the original source

R18 ICAO: CORSIA fuels life-cycle emissions

Fuel life-cycle methodology, including core emissions and land-use effects.

Read the original source

R19 LanzaJet (13 November 2025): Freedom Pines on-spec fuel announcement

Company announcement of on-spec jet fuel and full plant operation.

Read the original source

R20 Kumar et al. (2025): HTL of wastewater-grown algae; PNNL record

Energy Conversion and Management: X 27, 101096. DOI: 10.1016/j.ecmx.2025.101096.

Read the original source

R21 Zhou & Hu (2020): Catalytic Thermochemical Conversion of Algae and Upgrading of Algal Oil for the Production of High-Grade Liquid Fuel: A Review

Catalysts 10(2), 145; DOI: 10.3390/catal10020145. Conceptual conversion diagram, not proof of this strain's fuel yield.

Read the original source

R22 European Commission: biobased, biodegradable and compostable plastics

Definitions distinguish feedstock origin from biodegradation and composting conditions.

Read the original source

References continued

R23 Gupta & Dutra Molino et al. (2025): Engineering C. pacifica for high lipid and starch

ACS ES&T Engineering 5 (2025), 36–49. First online 25 November 2024. DOI 10.1021/acsestengg.4c00443. See Figs. 1 and 4–6.

Read the original source

R24 Dutra Molino et al. (2025): Description of C. pacifica

Algal Research 88 (2025), 104034. DOI 10.1016/j.algal.2025.104034. Species characterization and tolerance assays.

Read the original source

R25 Gupta, Oliver, Dutra Molino et al. (2025): endogenous transcription factors

Algal Research 90 (2025), 104192. DOI 10.1016/j.algal.2025.104192. See Figs. 5–6, assay methods and lipidomics.

Read the original source

R26 Dutra Molino et al. (2025): PHL7 secretion from C. reinhardtii

Scientific Reports 15 (2025), 24690. DOI 10.1038/s41598-025-09100-0. Substrate-specific enzyme and polymer assays.

Read the original source

R27 Diaz & Dutra Molino et al. (2025): PHL7 production in C. pacifica

bioRxiv preprint v1, 15 May 2025. DOI 10.1101/2025.05.14.654053. Three 80-L raceways described in the abstract; the separate numerical pond maxima remain unverified in the audit.

Read the original source

R28 Authors' supplementary dataset (2025)

Supplementary dataset, published 14 May 2025. DOI 10.5281/zenodo.15411934. Associated with R27; data were not independently recomputed in the audit.

Read the original source

R29 Gunawan et al. (2020): biodegradation of renewable PU foams

Bioresource Technology Reports 11 (2020), 100513. DOI 10.1016/j.biteb.2020.100513. Formulation-specific soil and compost tests.

Read the original source

R32 ADEME: short flight

225 g CO_2e/passenger-km; 300–1,000 km flights; includes contrail effects, excludes infrastructure.

Read the original source

R37 Algae Farm: current teaching game

Teaching simulation; resource/score rules are not validated process or LCA coefficients.

Read the original source